INHERITED FROM RADICALS

Rational exponents inherit the domain restrictions of radicals. Whether a negative base is allowed depends on the denominator of the exponent.

RATIONAL EXPONENT

a

split by **whether the denominator n is even or odd**

≥0

CASE 1

n even

Base must be non-negative

no real number squares to a negative number; the expression would require complex numbers

E.G. (-4) is not real

1/2

ℝ

CASE 2

n odd

Any real base

negative bases have real odd roots

E.G. (-8) $= -2$

1/3

ONE MORE RESTRICTION

When the exponent is negative, $a \neq 0$ is required as well: division by zero is undefined. The restrictions apply equally to and a .

$\sqrt[n]{}$ $1/n$